Interferon-α/β–Mediated Innate Immune Mechanisms in Dermatomyositis

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Dermatomyositis has been modeled as an autoimmune disease largely mediated by the adaptive immune system, including a local humorally mediated response with B and T helper cell muscle infiltration, antibody and complement-mediated injury of capillaries, and perifascicular atrophy of muscle fibers caused by ischemia. To further understand the pathophysiology of dermatomyositis, we used microarrays, computational methods, immunohistochemistry and electron microscopy to study muscle specimens from 67 patients, 54 with inflammatory myopathies, 14 with dermatomyositis. In dermatomyositis, genes induced by interferon-α/β were highly overexpressed, and immunohistochemistry for the interferon-α/β inducible protein MxA showed dense staining of perifascicular, and, sometimes all myofibers in 8/14 patients and on capillaries in 13/14 patients. Of 36 patients with other inflammatory myopathies, 1 patient had faint MxA staining of myofibers and 3 of capillaries. Plasmacytoid dendritic cells, potent $CD4^+$ cellular sources of interferon-α, are present in substantial numbers in dermatomyositis and may account for most of the cells previously identified as T helper cells. In addition to an adaptive immune response, an innate immune response characterized by plasmacytoid dendritic cell infiltration and interferon-α/β inducible gene and protein expression may be an important part of the pathogenesis of dermatomyositis, as it appears to be in systemic lupus erythematosus.

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Dermatomyositis (DM) is an inflammatory myopathy (IM) characterized by subacute or chronic progressive proximal weakness, a characteristic skin rash, and muscle inflammation.[1] Pathological studies of muscle tissue from patients with DM have reported early histological abnormalities in capillaries,[2] infiltration of muscle by B cells and $CD4^+$ T cells, perifascicular atrophy, and tubuloreticular inclusions (TIs) in endothelial cells.[3] The disease has been hypothesized to result from sequential binding of antibodies to an endothelial autoantigen, activation of complement, and vascular inflammation, with perifascicular atrophy the result of ischemia to watershed areas in the vascular supply of muscle.[4] However, it is unknown whether complement is activated by antibody-dependent or -independent mechanisms, whether endothelial injury precedes or follows complement deposition, whether the perifascicular regions are truly watershed areas, or if the perifascicular atrophy itself is, in fact, caused by ischemia.[5] Perifascicular atrophy was not present in animal models of ischemic myopathy, which instead found selective vulnerability of the central portions of fascicles and sparing of the perifascicular regions.[6,7]

To further understand the pathophysiological mechanisms of DM in muscle, we measured and analyzed large-scale gene transcriptional patterns in muscle tissue from patients with DM and compared these findings with patients with other inflammatory myopathies and with patients without neuromuscular disorders. These results led us to reevaluate previous characterizations of the cellular infiltrates and mechanisms involved in the pathogenesis of DM.

Patients and Methods

Patients

We performed microarray experiments on muscle samples collected prospectively from 54 patients with inflammatory myopathies (14 with DM, 20 with inclusion body myositis

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Table 1. Characteristics of 14 Patients with Dermatomyositis

				Clinical Histopathology				MxA Immunohistochemistry		Laboratory Studies		
Patient No.	Age/ Sex	Duration of Muscle Weakness/ Corticosteroid Treatment before bx (weeks)	Muscle Biopsied/ MRC Grade	Inflammatory Infiltrate (− to +++)[a]	Perifascicular Atrophy (− to +++)	MAC+ Endothelium (− to +++)	EM Tubuloreticular Inclusions (− to +++)	MxA+ Myofibers (− or +)	MxA+ Capillaries (− or +)	CK (IU/L)	ANA ds-DNA RNP Jo-1	Associated Diseases
P4	36/F	2/0.3	Bic 4+	None	−	+	+	−	+	13,066	−	−
P9	73/M	4/no	Bic 5	+	−	+	−	−	+	687	1:2,560	−
P12	73/F	4/0.4	Bic 4	None	−	−	+	+	+	3,683	−	−
P19	39/M	8/4	Delt 4−	++	+	+	ND	+	+	6,030	1:1,280 DNA 1:80	SLE
P30	65/M	6/no	Vaslat 5	+	−	ND	+	−	+	1,945	− Jo-1 +	ILD
P39	57/F	4/3	Delt 4	++	+[b]	+	−	−[b]	+	8,770	− Jo-1 +	ILD
P58	59/F	28/no	Bic 4+	++	++	ND	+	+	+	58	1:5,120	−
P86	69/F	6/6	Delt 4	None	+[b]	ND	ND	−[b]	−[b]	405	1:2,560 DNA - RNP 184	ILD MCTD
P87	33/F	12/no	Bic 4	++	+	ND	ND	+	+	179	1:10,240 DNA 588 RNP 167	MCTD
P168	59/F	6/0.1	Delt 4	None	−	+	−	−	+	2,364	−	−
P177	60/F	12/no	Bic 4	+	+	ND	ND	+	+	11,876	1:160	−
P179	22/F	14/no	Bic 4−	++	++	−	−	+	+	4,300	1:2,560	−
P181	50/M	12/no	Delt 1	+++	+++	ND	ND	+	+	7,867	1:2,560	−
P195	69/F	52/50	Bic 4	+	++	ND	ND	+	+	22	1:640	−

All patients except P86 had a skin rash.

Bx = biopsy; Bic = biceps brachii, delt = deltoid, vaslat = vastus lateralis; MAC = membrane attack complex; EM = electron microscopy; ND = not done; ILD = interstitial lung disease; ANA = antinuclear antibody; CK = creatine kinase; MRC = Medical Research Council; RNP = antiribonucleoprotein antibodies; normal <20 EU/ml; dsDNA or DNA-anti–double = stranded DNA antibodies, normal <25 IU; Jo-1 = anti-tRNA synthetase antibodies; M = male; F = female.

[a]Inflammatory infiltrates were graded based on review of routine hematoxylin and eosin and trichrome-stained sections. Because immunohistochemical stains are more sensitive, some patients without inflammatory cells seen on routine sections had inflammatory cells identified by immunohistochemistry.

[b]No perifascicular atrophy was present on MxA-stained sections but was present in hematoxylin and eosin–stained sections.

[IBM], 6 with polymyositis [PM], 5 with necrotizing myopathy, and 9 with myopathies with inflammation and of uncertain classification), 10 patients without clinical or histological evidence of a neuromuscular disorder (normal subjects), and 3 patients with inherited myopathies (1 suspected and 1 genetically confirmed with myotonic dystrophy type 2, and 1 with genetically confirmed facioscapulohumeral muscular dystrophy). Muscle samples from a total of 67 patients were studied with microarrays. Routine clinical histochemistry was performed on all of these specimens, and further detailed immunohistochemical studies were performed on 53 samples and on muscle samples from 7 additional patients not studied using microarrays who had unclassified myopathies with inflammation (n = 3), muscular dystrophies (n = 3; 2 with clinically defined limb-girdle muscular dystrophies, and 1 with genetically confirmed Becker muscular dystrophy), and central core disease (n = 1). Patients met criteria[8,9] for definite DM (n = 9), probable DM (n = 5), definite PM (n = 6), necrotizing myopathy (n = 5), or definite or possible IBM (n = 20). All patients were seen and examined by one of the study investigators. The clinical characteristics of patients with DM are listed in Table 1. Of the 14 patients with DM, 7 were treated with corticosteroids for a median duration of 12 days (range, 1–350 days) and 7 had never received immunosuppressive therapy. Biopsies were performed for clinical indications independent of this study. An institutional review board approved the study.

Microarray Studies

Microarray studies of muscle were performed as described previously,[10] except that Affymetrix U133A microarrays (Affymetrix, Santa Clara, CA) with 22,283 probe sets representing approximately 16,000 genes and Microarray Analysis Suite 5.0 (Affymetrix) were used in this study. Data underwent normalization by linear regression to a reference sample (P179). Microarray performance was unsatisfactory for 1 of 67 samples (P181 with DM). We were specifically interested in the most highly differentially expressed genes and used greater fold ratio cutoffs than most microarray studies. Marked overexpression of a gene in DM compared with normal was defined as a detection value of "Present" in at least 75% of patients with DM and a fold ratio of greater than 10 compared with normal. To determine genes with expression that was substantially increased in DM compared with both normal subjects and other patients with IM, we required a given gene detection value of "Present" in at least 75% of DM samples, a mean fold ratio in DM of at least 3.0 compared with normal, and a mean fold ratio in DM of at least 2.0 compared with the other IM samples. Substantial downregulation of a gene was analogously defined by a mean fold

Table 2. Monoclonal Mouse Anti–Human Antibodies Used in This Study

Antibody	Clone	Specificity	Source	Dilution	Incubation (hr)
MxA	M143	MxA protein	OH	1:1000	15
IFN-α	MMHA-2	IFN-α subtypes	PBL	1:100	15
CD3	T3-4B5	T cells	DK	1:300	1
CD4	MT310	T-H,MAC,MDC,PDC	DK	1:20	1
CD83	HB15a	Dendritic cells	BC	1:300	1.5
CD123	7G3	Plasmacytoid dendritic cells; Endothelial cells	BD	1:150	1.5
BDCA-2	AC144	Plasmacytoid dendritic cells	MT	1:10	5

IFN = interferon; T-H = T helper cell; MAC = macrophage; MDC = myeloid dendritic cell; PDC = plasmacytoid dendritic cell; Sources: OH = courtesy of Dr. Otto Haller, Department of Virology, University of Freiburg, Germany; PBL = PBL Biomedical Laboratories (Piscataway, NJ); DK = DakoCytomation; BD = BD Biosciences Pharmingen (San Diego, CA); BC = Beckman Coulter, (Miami, FL); MT = Miltenyi Biotec (Gladbach, Germany).

ratio less than 0.33 and 0.50 compared with normal subjects and other patients with IM, respectively. p values were determined using a two-tailed t distribution assuming equal variances. Genes were identified as interferon-α/β induced through searches of literature and molecular databases. For cluster analysis of genes and arrays, probe sets with absent detection values across all DM and normal samples were removed, and the remaining probe sets were filtered for a standard deviation of greater than 2,000. Data were mean centered and normalized, and then hierarchical average linkage clustering was performed using centered correlation coefficients, with software Cluster and TreeView (Eisen Lab, Berkeley, CA). All DM and normal microarray data are available via the internet at http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE1551.

Computational Gene Promoter Regulatory Element Identification

To further characterize the regulation of upregulated genes in DM by interferon-α, -β, and -γ, we retrieved genomic DNA sequences for all genes represented on the U133A microarray from the Ensembl human genome database version 16.33.1. For each such gene (N = 15,534), we searched for previously reported interferon-regulated gene promoter sites, the interferon-α/β–stimulated regulatory element, and the interferon-γ activation sequences.[11] We computationally searched 1,000bp upstream of the transcription initiation site for each gene on both DNA strands and computed the number of such sequence matches for each gene. p values were computed for the likelihood that a given number of occurrences of flanking various interferon-α/β–stimulated regulatory element and interferon-γ activation sequences in our list of differentially expressed genes had arisen by chance, comparing the number found with such occurrences within the remaining set of genes not highly upregulated, by summing up all the probabilities for the observed or greater number of occurrences using the hypergeometric distribution.

Immunohistochemistry

Mouse anti–human monoclonal antibodies were used (Table 2) for immunohistochemical studies. Cryostat muscle sections (10μm) were fixed in cold acetone or cold 1% paraformaldehyde, followed by soaking in cold 0.05mol/L tris(hydroxymethyl)aminomethane (Tris) buffer, pH 7.6, for 5 to 10 minutes. Sections fixed in 1% paraformaldehyde were soaked further in Tris buffer at room temperature. Slides were placed in Tris buffer supplemented with 3% porcine serum for 10 minutes before incubation with the primary antibodies; sections then were washed with Tris saline, soaked in the same Tris buffer, and incubated with peroxidase-labeled polymer conjugated to goat anti–mouse immunoglobulin antibody (EnVision+; DakoCytomation, Carpinteria, CA; or PowerVision; ImmunoVision Technologies, Daly City, CA) for 30 minutes, and then washed with Tris saline. Antibody localization was effected by using a peroxidase reaction with 3,3′-diaminobenzidine tetrahydrochloride (DAB+; DakoCytomation) as the chromogen. The staining reaction was intensified by using 3,3′-diaminobenzidine tetrahydrochloride–enhancing solution (Zymed Laboratories, South San Francisco, CA). Slides were washed with water, counterstained with methyl green, dehydrated, and mounted with Poly Mount (Poly Scientific, Bay Shore, NY).

Quantitative Analysis of CD3-, CD4- and BDCA2- Expressing Cells

Quantitative analysis of cells expressing CD3, CD4 or BDCA2 was performed on muscle samples from five patients with DM who had never been treated with immunosuppressive agents and from five normal control subjects. Each section was viewed using an Olympus BX41 microscope (Olympus, Melville, NY), and cells were counted at either ×40 (when a sparse number of cells were present) or ×100 (when dense infiltrates were present) magnification, with the aid of an Olympus DP70 10 megapixel camera (Olympus), digital image capture software (Olympus), and digital image–processing software (Adobe Photoshop; Adobe Systems, San Jose, CA). Digital magnification to an equivalent optical magnification of approximately ×600 was used as required. For each slide, we counted the total number of cells present in the entire section of muscle present and calculated the total cross-sectional area of each section using imaging software. Each cell was classified as perivascular, perimysial, or endomysial using the same criteria as Arahata and Engel[3]; although unlike that previous study, we looked at the total number of cells in the entire section, rather than three or four sites of cell accumulation. For each cell surface marker and in each patient specimen, the density of the marker was defined as the number of positively stained cells divided by the cross-sectional area of the section, and densities were computed separately for perivascular, perimysial, and endomysial populations, and the total number of cells.

Immunoelectron Microscopy

Biopsy specimens were fixed with 1% paraformaldehyde for 5 minutes, washed with phosphate-buffered saline (PBS) containing 0.2M glycine, infiltrated with 2.3M sucrose in PBS for 15 minutes, frozen at −120°C, sectioned, transferred to formvar-carbon–coated copper grids, and floated on PBS. Anti–myxovirus resistance 1 (MxA) antibodies and protein A gold were diluted with 1% bovine serum albumin and centrifuged for 1 minute at 14,000*g* before labeling to avoid possible aggregates. Grids were floated on drops of 1% bovine serum albumin for 10 minutes to block nonspecific labeling, transferred to 5μl of primary antibody, incubated for 30 minutes, washed in 4 drops of PBS for 15 minutes, transferred to 5μl drops of bridging antibody (rabbit anti–mouse; Dako, Carpinteria, CA) for 30 minutes, washed in PBS for 15 minutes, transferred to 5μl protein A gold (purchased from G. Posthuma, University Medical Center Utrecht, Utrecht, the Netherlands) for 20 minutes, and washed in 4 drops of PBS for 15 minutes and 6 drops of double-distilled water.

Contrasting and embedding of the labeled grids was performed on ice in 0.3% uranyl acetate in 2% methyl cellulose for 10 minutes, and grids were examined in a JEOL 1200EX transmission electron microscope (JEOL USA, Peabody, MA) at a primary magnification of ×20,000 to ×30,000.

Reverse Transcription Quantitative Polymerase Chain Reaction

Reverse transcription quantitative PCR for interferon-α–inducible genes (*MxA*, *G1P2*/*ISG15*, *CIG5*, *LY6E*, *ISG20*, *IFIT2*, and *OAS1*) was performed on three muscle samples (P83-IBM, P87-DM, and P128-Norm) using primers designed with Primer3 software (Whitehead Institute, Cambridge, MA) and purchased commercially (MWG Biotech, Ebersberg, Germany). Primers used were as follows: *MxA*: (forward) 5′-ACCTGATGGCCTATCACCAG-3′, (reverse) 5′-TTCAGGAGCCAGCTGTAGGT-3′; *CIG5*: (forward) 5′-TGCTTGCTTTCTCTGAGCTG-3′, (reverse) 5′-AACCTTGGGCAAGGAAGAAT-3′; *OAS1*: (forward) 5′-CAAGCTCAAGAGCCTCATCC-3′, (reverse) TGGGCTGTGTTGAAATGTGT-3′; *LY6E*: (forward) 5′-TGATGTGCTTCTCCTGCTTG-3′, (reverse) 5′-ACAGGTCTTGCTCAGGCTGT-3′; *ISG20*: (forward) 5′-AGATCCTGCAGCTCCTGAAA-3′, (reverse) 5′-TGTTCTGGATGCTCTTGTGC-3′; *IFIT2*: (forward) 5′-GCGTGAAGAAGGTGAAGAGG-3′, (reverse) 5′-GCAGGTAGGCATTGTTTGGT-3′; *ISG15*: (forward) 5′-TGTCGGTGTCAGAGCTGAAG-3′, (reverse) 5′-GCCCTTGTTATTCCTCACCA-3′.

RNA (1μg) was reverse-transcribed to complementary DNA using reverse transcription reagent kits (Roche Molecular Systems, Branchburg, NJ). The resulting complementary DNA was amplified in a real-time quantitative PCR using the *TaqMan* Universal PCR Master Mix, the recommended *TaqMan* protocol, and the ABI 7700 Sequence Detection System (Applied Biosystems, Foster City, CA). Standard curves for glyceraldehyde phosphate dehydrogenase were generated by plotting the log of template amount versus C(t) for a specific template. The outcome of each amplification run was compared with the standard curve as described by the *TaqMan* protocol (Applied Biosystems), enabling the calculation of the RNA expression level fold changes among the tissue samples.

Results

Interferon-α/β–Inducible Gene Transcripts Are Substantially More Abundant in Dermatomyositis Muscle Than in Other Inflammatory Myopathies

Clustering of 13 DM and 10 normal samples resulted in partial separation of these two classes and clusters of genes including some regulated by interferon-α/β (Fig 1). Of the most highly differentially upregulated genes, 12 of 14 (86%) are induced by interferon-α and -β. Reverse transcription quantitative PCR studies showed similar results (see supplementary Table 1). No genes were markedly downregulated. This pattern of gene expression was not present in other IMs, which had 3 of 21 (15%) highly upregulated genes inducible by interferon-α/β (data not shown). Two previous microarray studies in small numbers of patients also have found high expression of interferon-α/β–inducible gene transcripts in adult[10] or childhood[12] DM compared with normal subjects.

To identify characteristics of muscle inflammation specific to DM, we compared gene expression profiles of 13 patients with DM with 40 patients with other IMs, including IBM (n = 20), PM (n = 6), necrotizing myopathy (n = 5), and unclassifiable myopathies with inflammation (n = 9). Twenty-five genes were highly upregulated in DM compared with both normal subjects and other patients with IMs, of which 21 (84%) had been previously identified as inducible by interferon-α/β or had upstream sequence regulatory elements characteristic of such (Table 3). In contrast, similar analysis for genes upregulated in other IMs compared with DM produced 2 of 52 (4%) interferon-α/β–inducible genes. These findings demonstrate substantially greater interferon-α/β–induced gene expression in DM than in other IMs.

To further characterize the potential regulation by interferons of these differentially overexpressed genes, we computationally identified interferon-α/β–inducible gene promoter sequences associated with interferons and found them to be overrepresented in the set of DM highly upregulated genes (see supplementary Table 2). We identified one or more promoter sites in 18 of 25 (72%) genes (see supplementary Table 3). Only 1 of 25 genes (4%) had an interferon-γ promoter site. Genes upregulated in DM are almost exactly those identified as regulated by interferon-α/β to a much greater extent than interferon-γ in previous microarray studies of fibro-

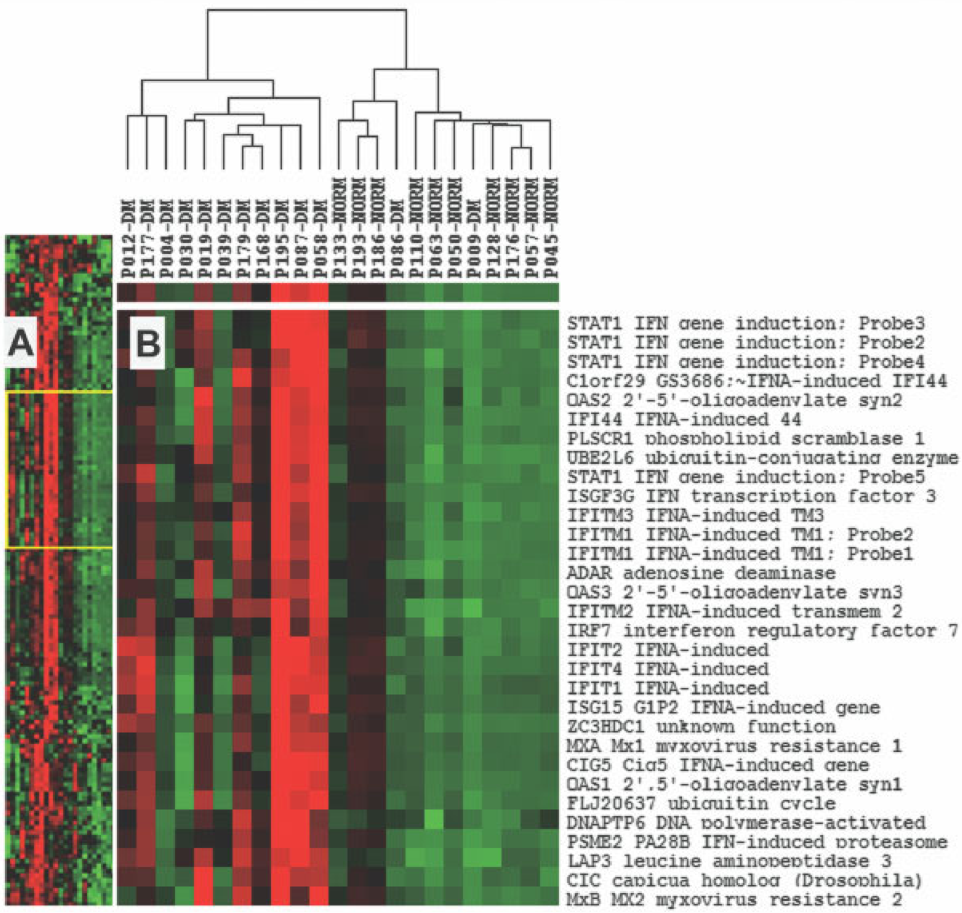

Fig 1. Cluster analysis of microarray data for dermatomyositis (N = 13) and normal (N = 10) muscle specimens. Hierarchical clustering performed as described in Patients and Methods. *(A) Overall image for 2,432 probesets in rows and 23 tissue samples in columns. (B) Enlargement of gene cluster, boxed in (a), showing a cluster of interferon-α/β (IFNA)–inducible genes, overexpressed in dermatomyositis compared with normals. Red indicates high expression; black indicates intermediate expression and green indicates low expression.*

blast in vitro differential responsiveness to interferon-α, -β, and -γ (see supplementary Table 4).[13] These findings provide additional evidence for a highly specific interferon-α/β response in DM.

Human Myxovirus Resistance 1 Protein Is Highly Expressed by Myofibers in a Preferentially Perifascicular Pattern and in Blood Vessels in Dermatomyositis but Not in Other Inflammatory Myopathies

The interferon-α/β–inducible human MxA protein was present by immunohistochemical staining in DM muscle with an intensity correlating with its microarray-measured transcript levels, which sharply differentiated most DM specimens from all of the other 50 samples studied (Fig 2). Of 14 DM samples stained for MxA protein (see Table 1), 8 showed expression by myofibers, either diffusely or in a preferentially perifascicular distribution that included both small ("atrophic") and normal size fibers, and 13 showed intense staining of capillaries and other blood vessels (Fig 3). When present, myofibers in regions of perifascicular atrophy always stained positively for MxA (seven samples). There was a weak association between duration of muscle weakness and the presence of MxA myofiber expression (median duration of 12 weeks in patients with MxA myofiber expression and 5 weeks in patients without MxA myofiber expression; $p = 0.02$). Of 46 other muscle samples (17 with IBM, 5 with PM, 1 with necrotizing myopathy, 8 with other IMs, 5 with dystrophies, and 10 normal subjects), only 1 showed expression of MxA by myofibers (2 groups of small atrophic fibers faintly staining in an IBM sample), and 3 showed expression of MxA by blood vessels (1 with central core disease, 1 with systemic lupus erythematosus (SLE) and myalgias but not myositis, and 1 with human immunodeficiency virus myositis). Notably, both SLE and human immunodeficiency virus

Table 3. Interferon-α/β–Inducible Genes Upregulated in Dermatomyositis Compared with Other Inflammatory Myopathies

Affymetrix ProbeSet	Gene Symbol	Gene Name	α/β-Interferon-Inducible by Literature/ISRE Promoter Site	Fold-DM-Other IM	DM-Norm Fold	DM-Other IM (*p*-value)	DM-Norm (*p*-value)
205483_s_at	GIP2	ISG15; interferon-stimulated gene 15	+/+	13.0	83.2	8.75E-08	3.56E-03
213797_at	CIG5	Viperin; IFNA-induced viral resistance gene	+/−	8.0	27.4	6.47E-07	5.47E-03
202086_at	MXA	Mx1; myxovirus resistance 1	+/+	7.3	13.6	2.60E-07	4.80E-03
205552_s_at	OAS1	2′,5′-oligoadenylate synthetase 1	+/+	5.9	12.8	1.16E-06	6.10E-03
204747_at	IFIT4	IFI-60/RIG-G; interferon-induced protein	+/−	5.6	21.4	3.69E-07	2.75E-03
204439_at	C1orf29	GS3686; homology to interferon-induced p44	+/+	4.6	11.9	1.18E-06	1.95E-03
203021_at	SLPI	Leukocyte protease inhibitor; IRF-1 induced	+/−	4.5	9.8	1.61E-04	3.04E-02
219211_at	USP18	IFNA-stimulated ISG43; ubiquitin protease	+/+	4.1	10.4	2.54E-08	7.58E-04
203153_at	IFIT1	ISG56; interferon-induced protein	+/+	4.0	4.7	3.96E-07	5.53E-03
204415_at	G1P3	Interferon, α-inducible protein (IFI-6-16)	+/+	3.8	9.5	1.31E-04	2.02E-02
200862_at	DHCR24	24-dehydrocholesterol reductase	−/−	3.6	3.8	2.06E-06	1.13E-02
214453_s_at	IFI44	IFNA-induced p44 (hepatitis C microtubular)	+/+	3.5	10.8	1.04E-06	1.07E-03
218400_at	OAS3	2′-5′-oligoadenylate synthetase 3	+/+	3.2	5.1	2.09E-07	9.15E-04
208436_s_at	IRF7	Interferon regulatory factor 7	+/+	3.1	7.1	8.40E-09	1.33E-04
205047_s_at	ASNS	Asparagine synthetase	−/−	2.9	6.9	2.10E-05	4.68E-03
204972_at	OAS2	2′-5′-oligoadenylate synthetase 2	+/+	2.9	4.8	5.56E-06	3.93E-03
219863_at	CEB1	Cyclin E binding protein 1	+/+	2.8	4.7	6.90E-07	1.74E-03
219209_at	MDA5	Melanoma differentiation associated protein-5	+/+	2.6	13.7	7.31E-05	2.09E-03
221232_s_at	ANKRD2	Ankyrin repeat domain 2	−/−	2.6	3.6	1.29E-05	2.61E-03
204994_at	MXB	Mx2; Myxovirus resistance 2	+/+	2.5	4.4	2.68E-05	4.63E-03
202446_s_at	PLSCR1	Phospholipid scramblase 1	+/+	2.5	5.9	1.29E-05	1.20E-03
218986_s_at	NM017631	Hypothetical protein FLJ20035	−/+	2.4	6.3	7.59E-05	2.29E-03
206133_at	XAF1	XIAP associated factor–1	+/+	2.2	3.4	1.01E-03	1.88E-02
202411_at	IFI27	IFI27 interferon, α-inducible protein 27	+/+	2.1	9.5	3.67E-03	3.70E-03
218145_at	TRB3	TRB3/SKP3	−/−	2.1	4.6	2.48E-02	7.29E-02

Highest differentially upregulated genes in dermatomyositis (N = 13) compared with other inflammatory myopathies (N = 40; no patients had SLE or MCTD) and nonneuromuscular disease controls (N = 10) from a total of 22,283 probesets. Twenty-five genes met the criteria for expression in DM compared with normal with a fold-ratio of ≥3.0, "Present" in at least 75% of DM samples, and additional upregulation with a fold ratio of ≥2.0 compared with average expression levels of all other inflammatory myopathy samples. *p* values for the distribution of expression levels in DM compared with other IMs are given and compared with normal. Twenty-one of the 25 genes (84%) are interferon-α/β inducible, according to either prior reports in the biomedical literature or the presence of an interferon-α/β gene promoter site (ISRE) identified in our studies (see supplementary Table 3).
DM = dermatomyositis; ISRE = interferon-stimulated regulatory element; IM = inflammatory myopathy.

are associated with excess interferon-α and MxA protein overexpression in serum or tissues.[14,15] Inflammatory cells were frequently MxA$^+$ in all diseases.

Myxovirus Resistance 1 Blood Vessel Reactivity in Dermatomyositis Localizes to Endothelial Cell Cytoplasmic Inclusions

We performed immunoelectron microscopy to localize MxA immunoreactivity in three DM muscle specimens (P58, P179, and P195) and five normal control samples and found gold-labeled anti-MxA antibody localizing within DM muscle endothelial cells in two samples, typically accumulating in cytoplasmic electron-dense bodies (Fig 4). Background deposition of gold particles was variably seen in normal control muscle, but no accumulations or electron dense bodies were seen.

These electron-dense, cytoplasmic, MxA-containing bodies could be TIs, a pathological feature of DM. TIs are tubular structures with a cross section of 25 to 28nm that localize to smooth endoplasmic reticulum, contain ribonucleoprotein, and form in endothelial and other cell types exposed to interferon-α in vitro.[16] Because MxA self-oligomerizes into 25- to 35nm tubular

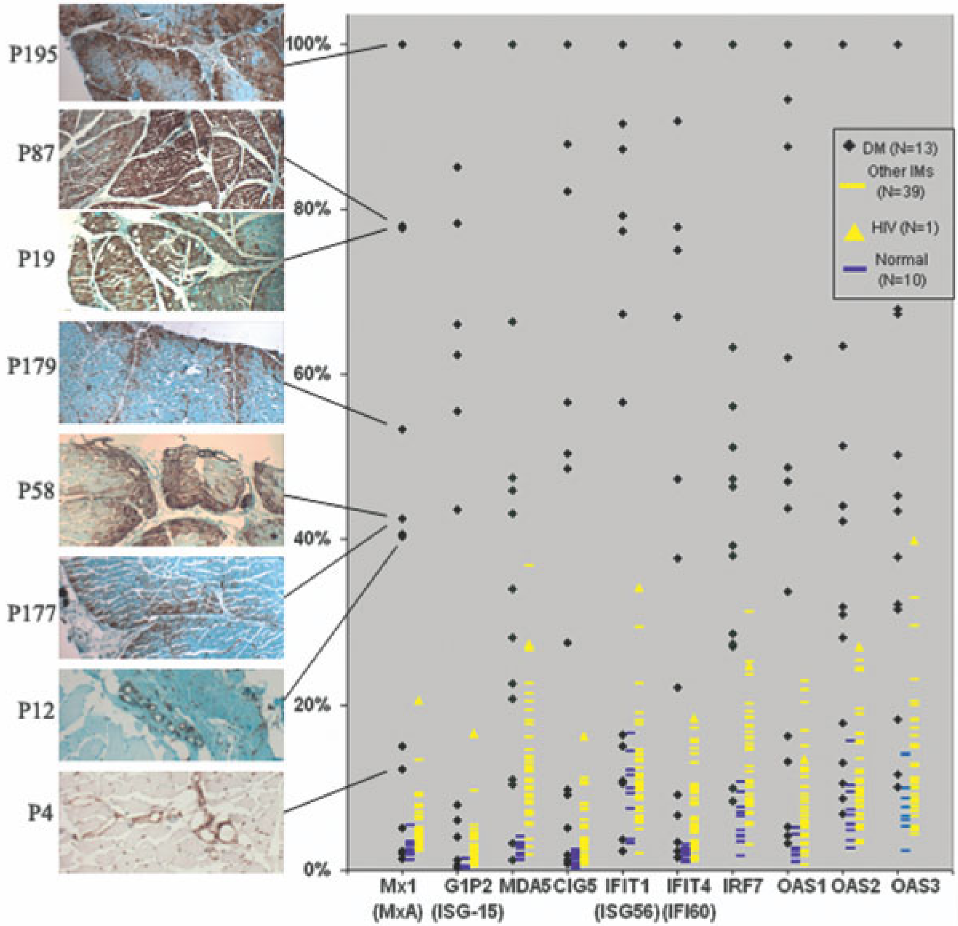

Fig 2. Microarray measured expression levels of 10 interferon-inducible genes for 63 tissue samples. Normalized expression levels are given as a percentage of the maximum value per gene for each of 10 genes across 63 samples (dermatomyositis; [DM], 13; normal, 10, and other inflammatory myopathies, 40). Immunohistochemistry for eight labeled specimens shown in left panel with each sample's corresponding MxA gene expression level. MxA transcript levels correlate well with tissue staining patterns; the highest levels are associated with intense myofiber staining (six tissue sections shown) and lower levels with staining limited to blood vessels (two tissue sections shown). Inclusion body myositis and polymyositis also have elevated amounts of interferon-α/β–inducible gene transcripts compared with normal, though they remain substantially lower than DM. A patient with HIV myositis had the highest levels among the other inflammatory myopathies, though remaining lower than most patients with DM.

structures within smooth endoplasmic reticulum,[17] binds ribonucleoprotein,[18] and is induced by interferon-α, MxA may be the principal constituent of TIs. Our immunoelectron microscopy technique did not preserve subcellular morphology sufficiently to identify TIs as they have been visualized previously in glutaraldehyde-fixed and plastic-embedded sections.

Plasmacytoid Dendritic Cells Are Abundant in Dermatomyositis

The pattern of interferon-inducible gene expression in DM led us to search for plasmacytoid dendritic cells (PDCs), which secrete interferon-α. These cells have been characterized previously in peripheral blood as $CD4^+CD68^+CD123^+BDCA\text{-}2^+$. PDCs identified as BDCA-2$^+$ cells were abundant in 10 of 14 patients with DM and were present in muscle sections that lacked inflammatory cells as seen in routine clinical histopathological studies. It is common to see inflammatory cells on immunohistochemical stains despite their absence on hematoxylin and eosin stains. Notably, the BDCA-2 marker is present only in a subset of PDCs; it is downregulated after maturation (in vitro exposure to interleukin-3).[19] Endomysial PDCs typically had dendritic elongated and branching morphologies (Fig 5), whereas perivascular and perimysial cells were often of plasmacytoid morphology with eccentric nuclei and abundant cytoplasm (Fig 6). Similar-appearing plasmacytoid cells stained positively for interferon-α (see Fig 6) or CD123 (Fig 7) in some cases. Quantitative studies (Table 4) of BDCA-2$^+$ cells showed that the density of BDCA-2$^+$ cells was a mean of 42% (range, 30–90%) that of CD4$^+$ cells. Because BDCA-2 is expressed only on a subset of PDCs, which

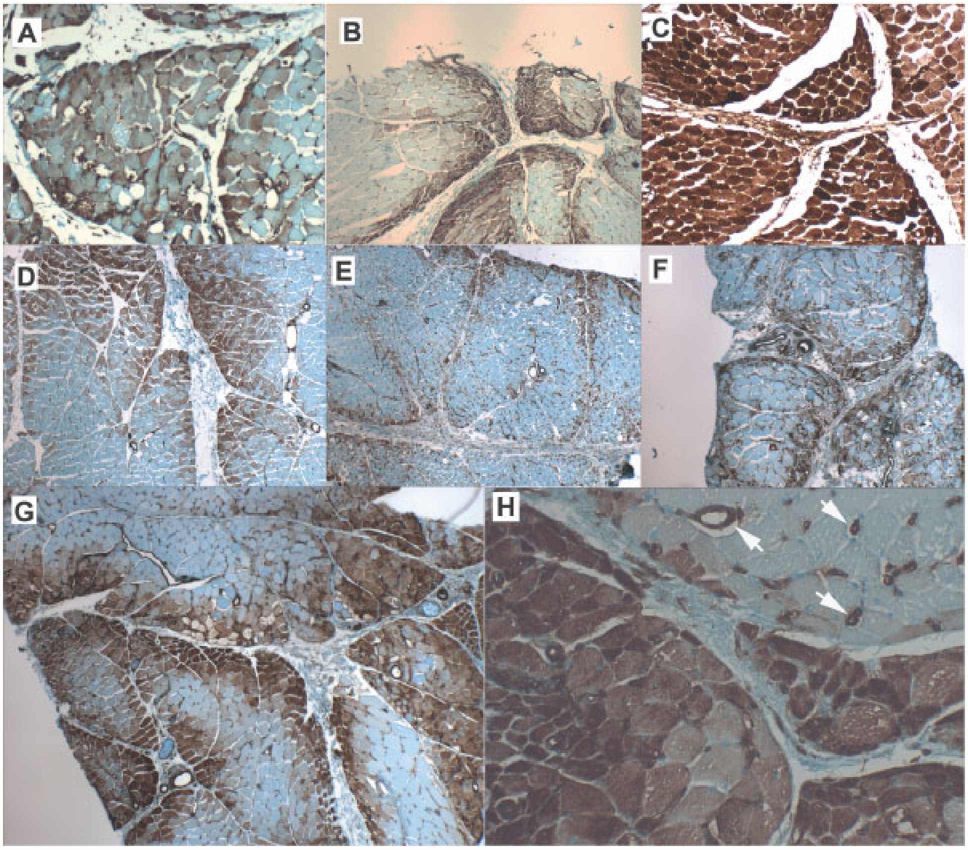

Fig 3. MxA protein immunohistochemistry in dermatomyositis. Patients (A) P19, (B) P58, (C) P87, (D) P177, (E) P179, (F) P181, and (G) P195. (H) P195 higher magnification, showing dense staining of perifascicular and central myofibers, and arterioles and capillaries (arrows). ×40 and ×100 magnifications.

downregulate BDCA-2 after in vitro maturation,[19] this density estimate is a lower limit for PDCs. Overall, these findings demonstrate that DM muscle is characterized by infiltration with large numbers of PDCs.

Most CD4$^+$ Cells in Muscle from Patients with Dermatomyositis Are Not T Cells

Because PDCs express CD4, we wondered whether these accounted for some of the previously seen CD4$^+$ cells in DM that have been interpreted as T helper cells. In 13 of 14 DM specimens, qualitatively greater numbers of cells expressed CD4 than CD3 in perivascular, perimysial, and endomysial sites, particularly in the latter (Fig 8). Quantitative studies of samples from five untreated patients with DM (see Table 4) showed a mean multiple of 5.3 (range, 1.4–9.5) times as many total CD4$^+$ as CD3$^+$ cells present in DM. These crude estimates suggest that PDCs account for at least 30 to 90% of CD4$^+$ cells, and that they outnumber CD4$^+$ T cells by at least a factor of 2.

Discussion

Autoimmunity generally has been conceived of as occurring when a "specific adaptive immune response is mounted against self antigens,"[20] although the possibility of inappropriately active innate immune dysfunction contributing to human disease has been considered.[21] For the inflammatory myopathies, adaptive immune mechanisms have been studied intensively, whereas less consideration has been given to the possibility that effector mechanisms of the innate immune system might play a role in tissue injury. Involvement of the complement system in DM, a component of the innate immune system, has been recognized, but whether it is humorally activated[4,22] or activated by antibody-independent mechanisms is uncertain. Our studies suggest a significant role in DM for PDCs, effector cells of the innate immune system,[23] and interferon-α/β–mediated innate autoimmunity.

Human MxA is an interferon-α/β–inducible protein providing innate defense against several RNA viruses,[18] including influenza, by interfering with cytoplasmic transport of viral nucleocapsids and viral assembly. Myofibers in regions of perifascicular atrophy in our sections always expressed MxA, and MxA also was present in DM normal-appearing perifascicular and sometimes central fibers. Furthermore, expression of interferon-α/β–inducible genes and expression of cap-

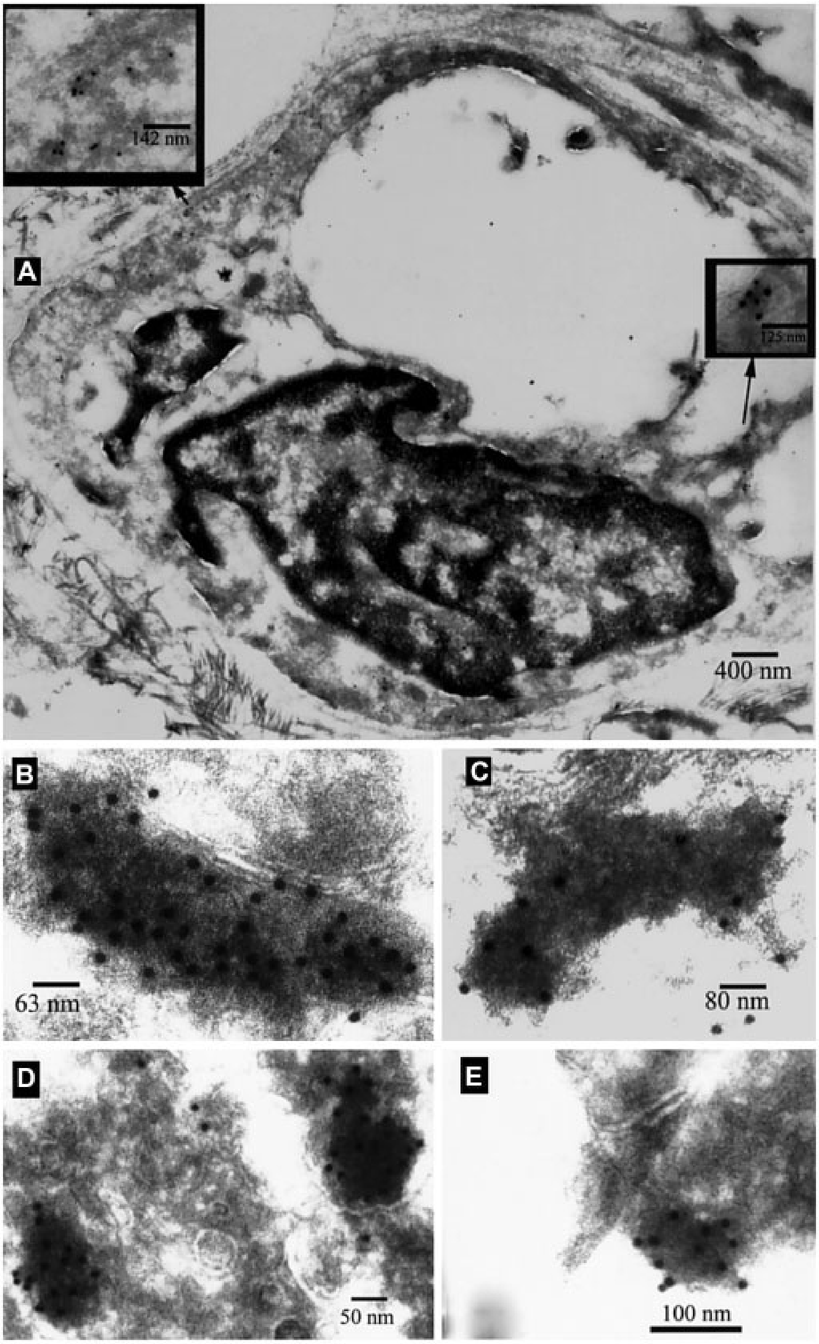

Fig 4. Immunoelectron microscopy for myxovirus resistance 1 (MxA) in two dermatomyositis (DM) samples. Gold-labeled anti–MxA antibody localizes within the cytoplasm of endothelial cells, sometimes to electron-dense cytoplasmic bodies of uncertain identity. (A) Sample P179, endothelial cell with cytoplasmic gold-labeling, enlarged in two insets. (B–E) Sample P195, higher magnification of cytoplasmic gold-labeled bodies from other endothelial cells. Magnification ×20,000–80,000.

illary MxA protein were early events in relation to muscle weakness and were present in 6 of 7 patients who had not received treatment at the time of biopsy.

MxA protein expression by myofibers in DM, in a preferentially perifascicular distribution, and within endothelial cells, together with its absence from myofibers and capillaries in IBM and PM, suggests that MxA and other interferon-α–inducible genes are important to the pathophysiology of DM. MxA is the only protein currently demonstrated to be specifically expressed by perifascicular fibers in DM but not by myofibers in other inflammatory myopathies or dystrophies. One study noted expression of the transcription factor signal transducer and activator of transcription 1 (STAT-1)

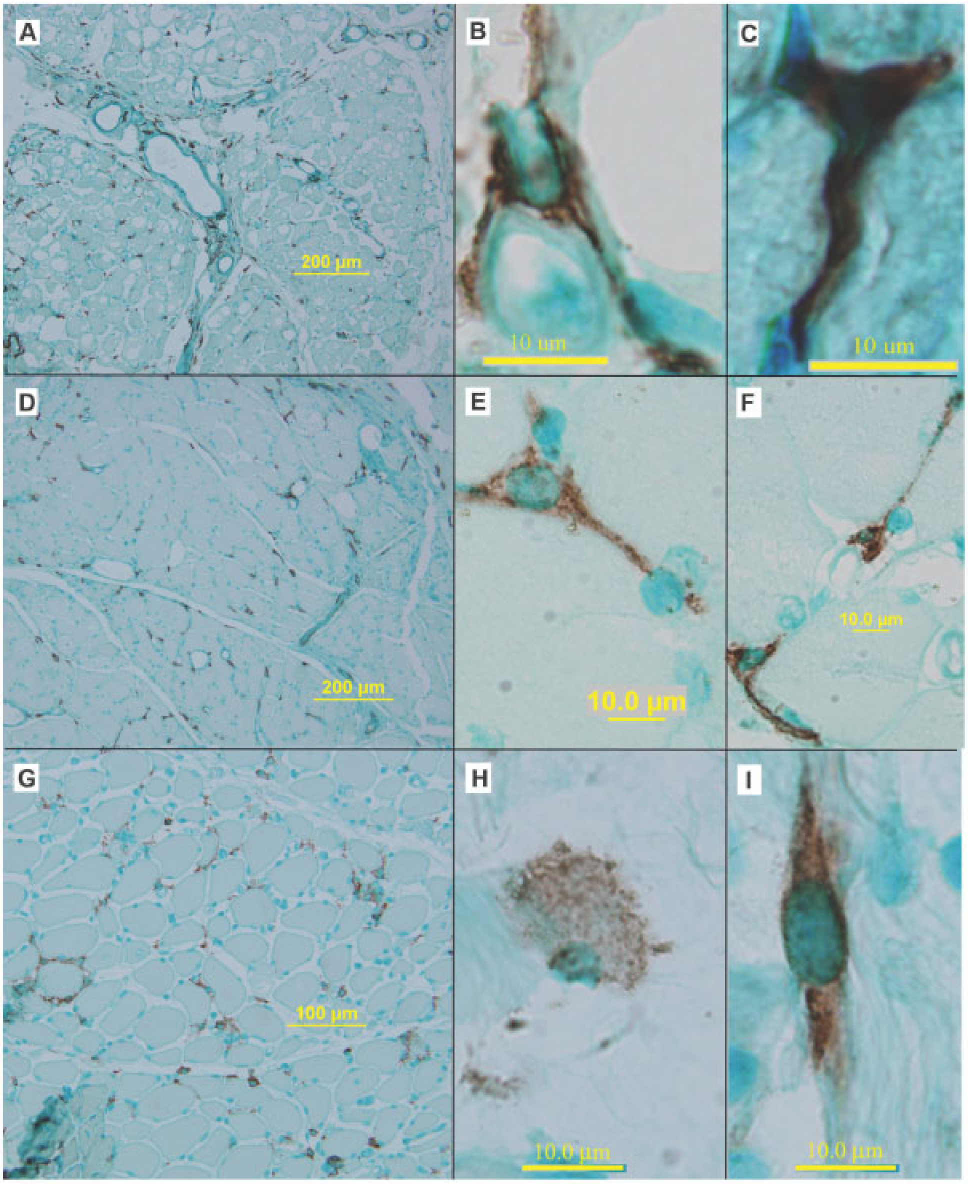

Fig 5. Plasmacytoid dendritic cells (PDCs) in dermatomyositis expressing BDCA-2. (A–C) P87-DM, (D–F) P195-DM, and (G–I) P179-DM. (A, D, G) Low-power views show PDCs are present diffusely throughout endomysial, perimysial, and perivascular regions. (B, C, E, F, I) High-power views of cells have dendritic (elongated or branching) morphology or (H) plasmacytoid form with eccentric nucleus and a large amount of cytoplasm.

protein in perifascicular regions in DM, although staining also was seen in some patients with PM, and IBM muscle was not studied.[24] The upregulation of STAT-1 was hypothesized to be secondary to interferon-γ. However, interferon-α/β also induces STAT-1, whereas MxA is not induced by interferon-γ (see supplementary Table 4).[13] Whether the expression of MxA or other interferon-inducible proteins serves as markers of diseased myofibers or mediate myofiber and endothelial cell injury is uncertain from our studies. However, their specific presence in DM and not IBM or PM, which are other disorders with myofiber injury, suggests that marked overexpression of one or more of these proteins may directly injure muscle fibers and endothelial cells. Previous immunohistochemical studies have shown that perifascicular myofibers in DM express a variety of other proteins, including major histocompatibility (MHC) class I molecules, STAT-1[24] (both interferon-α–inducible proteins), and αB-crystallin,[25] that could mediate myofiber damage in DM. However, unlike MHC class I expression (present on muscle fibers in other inflammatory myopathies[26] and dystrophies[26,27]), STAT-1 expression (present in some patients with PM and not studied in IBM[24]),

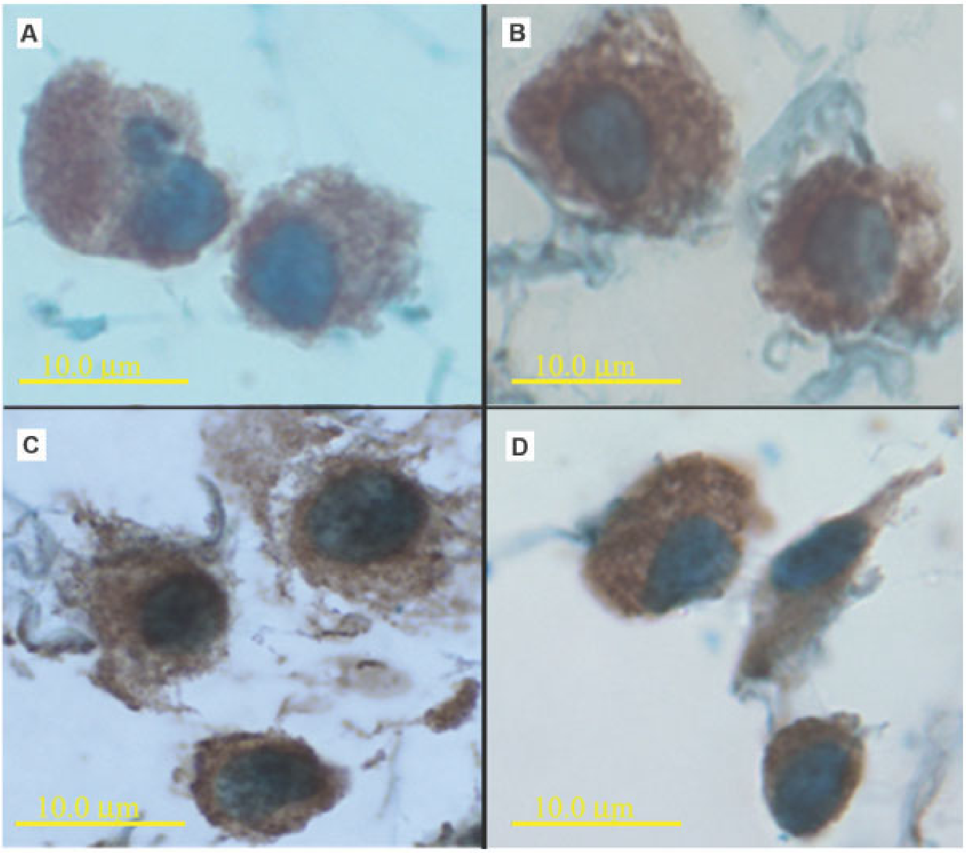

Fig 6. Plasmacytoid dendritic cells (PDCs) in dermatomyositis, in their plasmacytoid forms, expressing BDCA-2 and interferon-α. (A, B) BDCA-2 staining PDCs in their plasmacytoid form in (A) P12-DM and (B) P19-DM. Interferon-α staining PDCs in the same samples (C) P12-DM and (D) P19-DM.

and αB-crystallin (present in IBM, PM, myofibrillar myopathy, type II fiber atrophy, and denervation atrophy[25]), MxA expression in capillaries and myofibers was highly specific for DM. If overexpression of these other proteins (MHC class I molecules, STATs, and αB-crystallin) caused perifascicular atrophy, then perifascicular atrophy would be expected to be present in other disorders associated with their expression. Further independent confirmation of the specificity of MxA immunoreactivity in myopathies could clarify its relation to myofiber injury.

Our immunoelectron microscopy experiments provide limited but not conclusive evidence that MxA may be a component of the endothelial TIs characteristic of DM. This finding should be considered suggestive and requires further study to understand a possible relation between MxA and TIs.

Current models of DM propose that antibody-dependent, complement-mediated capillary injury is followed by perifascicular atrophy, the result of ischemia in a watershed distribution of blood flow in these regions. Microvascular injury and muscle infarction are definitively established pathogenic mechanisms in DM,[28] but it is unclear if perifascicular atrophy results from ischemia, as has been stated in review articles.[4] MxA and other interferon-α/β–inducible genes have been studied extensively for their antiviral roles, but they have not been reported as upregulated in ischemic tissues. In addition, the expression of MxA on both capillaries and muscle fibers in DM suggests that a common mechanism produces these abnormalities in both locations, rather than the capillary lesion resulting in perifascicular atrophy. Our data do not support a model of ischemia as the primary mechanism of perifascicular atrophy in DM.

Because we found a pattern of interferon-inducible gene expression in patients with DM, we searched for and found PDCs in their muscle samples. These cells, also known as natural interferon-producing cells, produce interferon-α in quantities 200 to 1,000 times greater than any other cell type.[23] Their presence in DM muscle suggests local cellular production of interferon-α/β. Some of our patients had minor PDC infiltrates, but marked upregulation of interferon-α/β–inducible genes and MxA protein expression, which strongly suggests that systemic production of

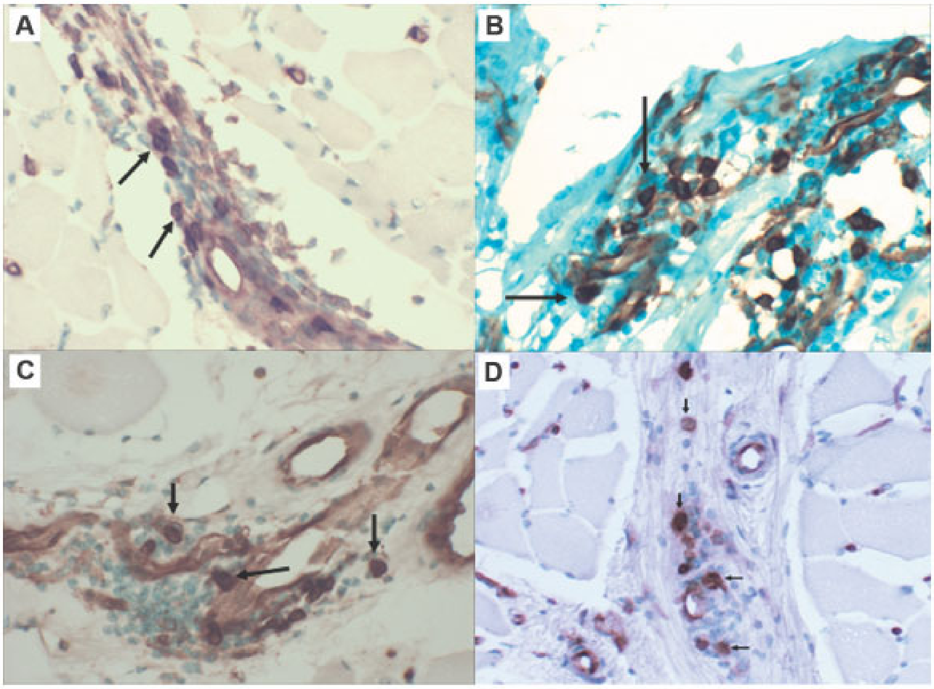

Fig 7. CD123⁺ plasmacytoid dendritic cells (arrows) in four patients with dermatomyositis. Patients (A) P87, (B) P58, (C) P19, and (D) P30. Magnification ×400.

interferon-α is likely important in DM as well. Interferon-α/β production in DM could result from the binding of short specific nucleotide sequences, CpG oligodeoxynucleotides, to Toll-like receptor 9 expressed on PDCs.[29]

Previous studies identify muscle infiltrates of $CD4^+$ cells in DM, PM, and IBM and interpret these as T helper cells.[3] In these prior studies, the $CD4^+$/total T cell ratio in the endomysium of DM muscle was found to be significantly greater than in IBM or PM. Our finding of PDCs,[30] $CD4^+$ cells that are not T cells or macrophages, could explain the greater relative numbers of endomysial $CD4^+$ cells previously seen in DM. Furthermore, the view that there are few inflammatory cells in the endomysium[28] is not supported by our findings. Although not present as concentrated accumulations of cells in DM, in our studies, the total number of endomysial cells exceeded the combined number of perimysial and perivascular cells. These cells are poorly seen on conventional trichrome and hematoxylin and eosin stains. Whereas DM previously has been modeled as a disease having selective enrichment in muscle of T helper cells and T helper cell–dependent stimulation of B cells,[4,28] we find relatively few T helper cells, many PDCs, and production of interferon-α–inducible transcripts and protein suggesting in part local innate immune effector mechanisms in addition to adaptive immune mechanisms.

Our findings, together with those of others, suggest that DM and SLE share a similar pathophysiology. Expression profiles of mononuclear blood cells from patients with SLE are similar to those we found in DM muscle (Table 5).[31–34] Interferon-α may mediate

Table 4. Quantitative Analysis of the Density of $CD3^+$, $CD4^+$, and BDCA-2^+ Cells in Untreated Patients with Dermatomyositis

Cells	Total (per mm^2)	Perivascular	Perimysial	Endomysial
CD3 (All T cells)	30 ± 23	19 ± 19	1 ± 1	10 ± 9
CD4 (T-H, MAC, MDC, PDC)	100 ± 89	22 ± 18	7 ± 3	71 ± 88
BDCA-2 (PDC subpopulation)	49 ± 33	12 ± 10	3 ± 2	33 ± 28

Patients with dermatomyositis (N = 5) who had never received immunomodulating therapies for their myopathy were studied. Density numbers represent the number of cells per square millimeter of cross-sectional area of a 10μm-thick section given as mean ± standard deviation.

T-H = T helper cell; MAC = macrophage; MDC = myeloid dendritic cell; PDC = plasmacytoid dendritic cell.

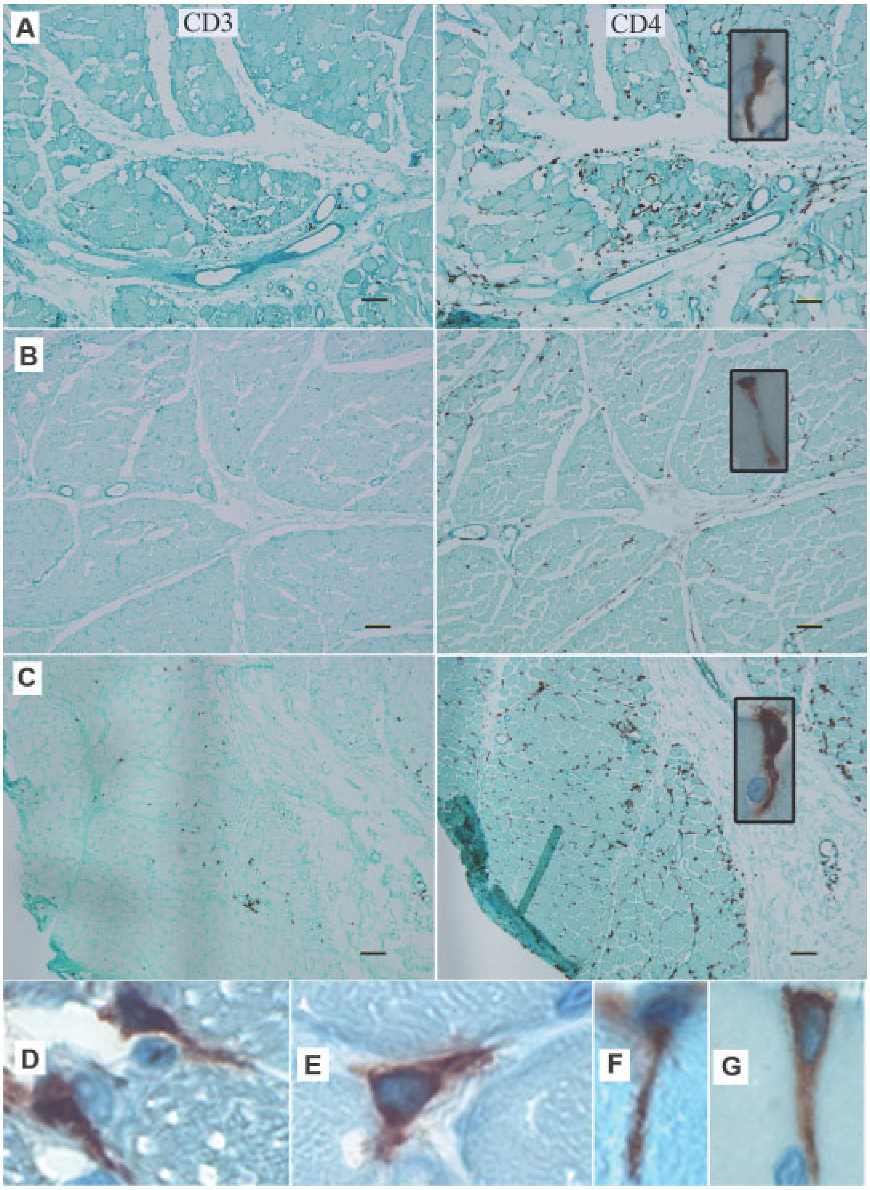

Fig 8. CD4 compared with CD3 expression and dendritic morphologies of CD4^{+} cells in seven patients with dermatomyositis. Greater numbers of CD4^{+} cells than CD3^{+} cells were seen in 13 of 14 patients with DM. (A–C) CD3 shown in left panels, CD4 in right panels, and CD4 high magnification in right panel insets showing elongated morphologies of cells. Patients (A) P19, (B) P87, and (C) P179. Higher magnification of elongated morphologies of CD4^{+} cells shown in (D) P86, (E) P168, (F) P177, and (G) P195. Scale bar = 100μm.

the pathogenesis of SLE,[35–37] and our studies suggest it may mediate the pathogenesis of DM as well. Increased serum levels of interferon-α and MxA are variously present in SLE and have correlated with clinical measures of disease activity.[38] The pathological findings in muscle between DM and SLE may be indistinguishable,[39] including perifascicular atrophy and the presence of TIs in endothelial cells. TIs were described in patients with SLE and were called lupus inclusions[40] before their recognition in patients with DM.[41] TIs correlate with increased serum interferon-α levels in patients with SLE and other disorders,[42,43] and they are associated with interferon-α treatment.[44] TIs develop in vascular endothelial cells exposed to interferon-α in vitro.[45] SLE sera induce lupus inclusions in vitro and stimulate PDCs to produce interferon-α.[46] In skin, membrane attack complex deposition, PDCs, and MxA protein occur in SLE.[14,47] Strategies to inhibit PDC interferon-α production, which are being considered

Table 5. Similar Gene Expression in Dermatomyositis and Systemic Lupus Erythematosus

Gene Symbol	Alternative Gene Names and Symbols	DM-Current Study Fold-Ratio	SLE-1	SLE-2	SLE-3	SLE-4
MxA	Mx1; Myxovirus resistance 1	13.6	+		+	
MxB	Mx2; Myxovirus resistance 2	4.4			+	
OAS1	OIAS	16.1		+	+	+
OAS2	p69OAS	4.8		+	+	+
OASL	p59OASL	2.3	+	+		+
IFIT1	ISG-56K/IFI-56/G10P1	4.7		+	+	+
IFIT2	IFI-54/G10P2	5.2		+	+	
IFIT4	IFI-60/RIG-G	21.4		+	+	+
IFI44	Hep C-microtubular p44	10.8			+	
G1P2	IFI-15/ISG15	83.2	+		+	
G1P3	IFI-6-16	9.5	+			
C1orf29	Homology to Hep C p44	11.9	+		+	
IRF7	Interferon regulatory factor 7	7.1			+	
LY6E	TSA1/RIG-E	10.1	+	+	+	+
PLSCR1	Phospholipid scramblase	5.9	+		+	
XAF1	XIAP associated factor-1	3.4	+		+	

Gene signature in common for DM muscle and SLE blood across five studies. Current data compared with four previously published studies of gene expression signatures in SLE blood. Fold-ratios compared with normal are given for the current study, and + indicates significant upregulation of a gene as determined by authors of the various other studies. The number of genes considered stringently differentially expressed in each of the studies were: DM-Current Study (25 genes), SLE-1[33] (161 genes), SLE-2[31] (61 genes), SLE-3[32] (15 genes), and SLE-4[34] (25 genes).
DM = dermatomyositis; SLE = systemic lupus erythematosus.

for treatment of SLE,[32,37] might be applicable to DM treatment as well.

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